

NACS 645 – Is face processing modular

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PROGRAM IN
NEUROSCIENCE &
COGNITIVE SCIENCE

Modularity of mind

Originates from Simon (1962?). Fodor (1983) distinguished modular and non-modular systems. Modular systems fulfill certain properties:

1. Domain specificity
2. Mandatory operation
3. Limited central accessibility
4. Fast processing
5. Informational encapsulation
6. 'Shallow' outputs
7. Fixed neural architecture
8. Characteristic and specific breakdown patterns
9. Characteristic ontogenetic pace and sequencing

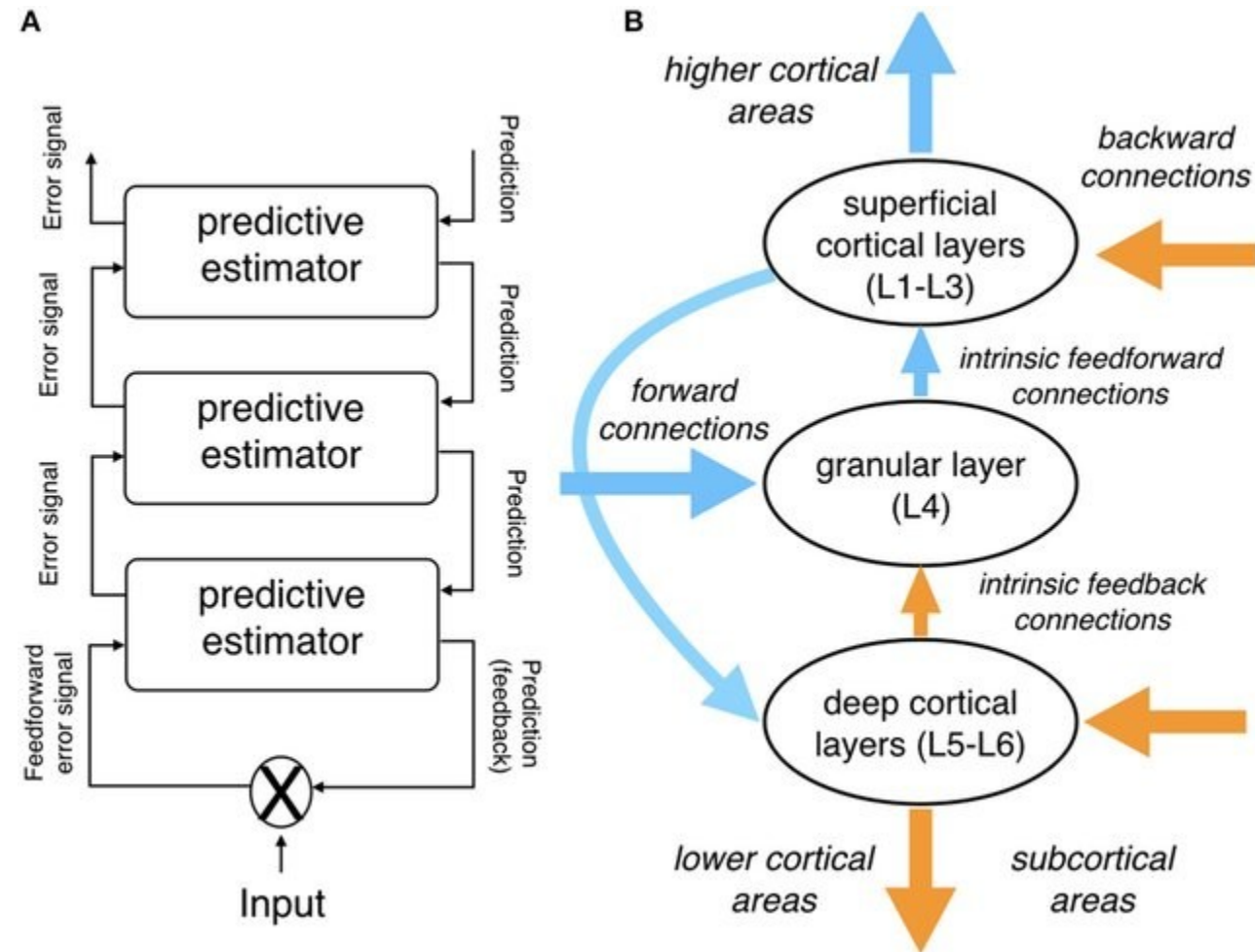
Modular systems are encapsulated : unaffected by other cognitive domains (cognitively impenetrable) and no information is added (non inferential).

Fodor's (1983) Modularity

1. **Domain specificity:** A modular system processes a **restricted class of inputs and problems**.
2. **Mandatory operation:** Once presented with the appropriate stimulus, **the system automatically engages and runs** to completion **without conscious control**.
3. **Limited central accessibility:** The intermediate **representations** generated within the system are **opaque to consciousness and cannot be accessed for explicit report**.
4. **Fast processing:** Modular systems **operate rapidly**, often on the order of hundreds of milliseconds, reflecting efficiency gained from restricted informational access.
5. **Informational encapsulation:** A module's **operations draw only on its proprietary database and input signals**, remaining insulated from information elsewhere in the mind.
6. **'Shallow' outputs:** The system produces relatively **simple, general, or basic-level representations that are computationally cheap**, rather than highly abstract or theoretical concepts.
7. **Fixed neural architecture:** Modular systems are **implemented in dedicated and circumscribed neural circuits** that can be selectively impaired without disrupting other functions.
8. **Characteristic and specific breakdown patterns:** **Damage** to a module **produces distinctive deficits** (e.g., prosopagnosia, dyslexia) **without widespread cognitive collapse**.
9. **Characteristic ontogenetic pace and sequencing:** Modules emerge and mature according to a species-typical developmental timetable, **largely triggered rather than learned**.

Predictive coding

- Brain as a **hierarchical prediction machine**
Hi-levels: prediction; Lo-levels: PE
- Continuous hypothesis testing:**
it infers causes of input and revises its model when errors appear (active inference)
- Neural architecture is hierarchical/layered**
(low → mid → high)
- Information flows top-down and bottom-up**
(continuous feedback loops)
- Priors, expectations, task demands determine** how strongly predictions **constrain** early processing
- Attention and uncertainty weight signals:**
precise prediction errors override priors, while imprecise ones are down-weighted

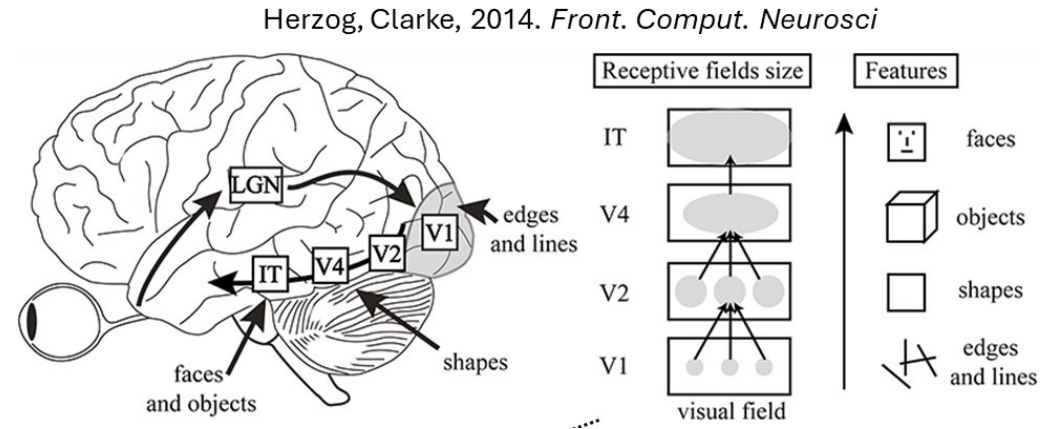


Functional Modularity (cog. psyc.)

The brain is hierarchically modular, with **modules** and **non-modules**.

Modules are defined by:

- **Fodor (1983)**: cognitive impenetrability, information encapsulation, innate modules

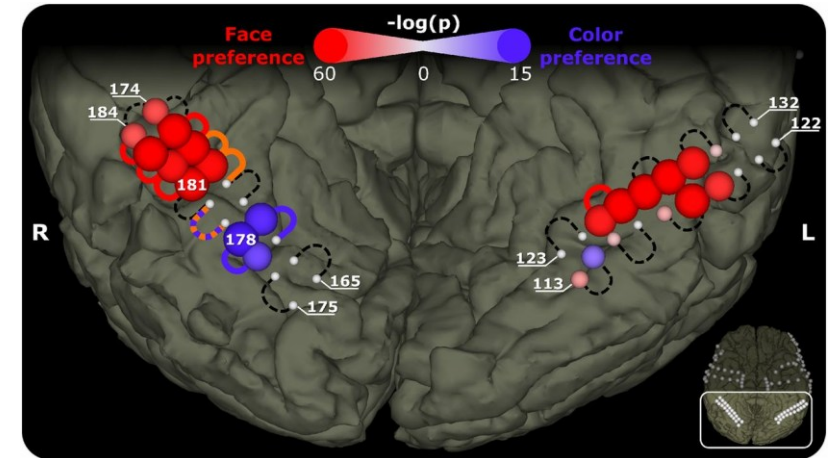


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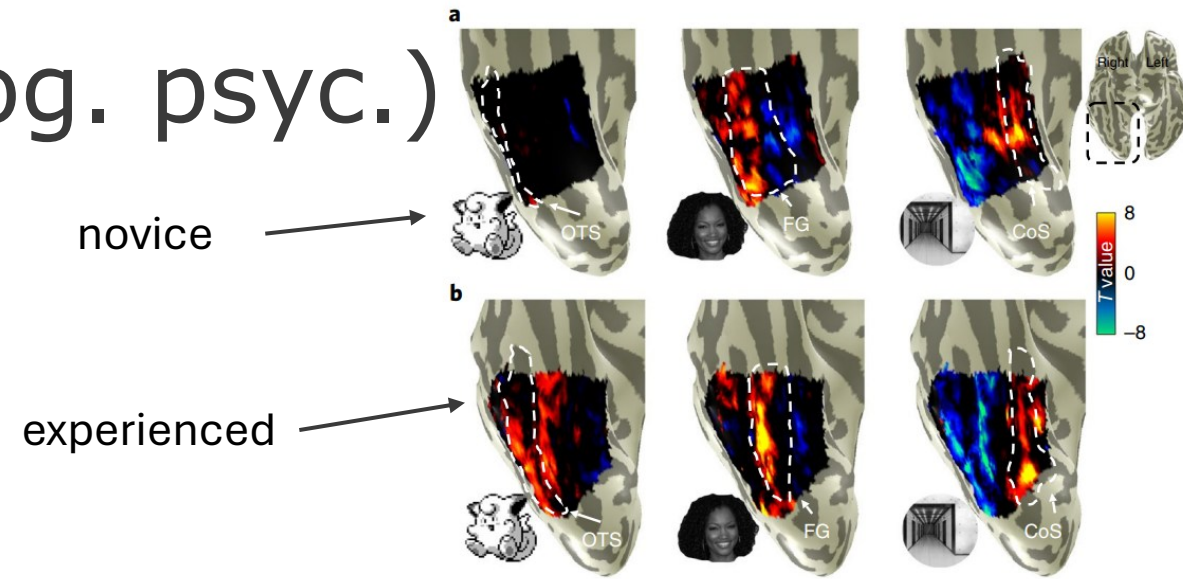


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Modules are defined in terms of (cognitive-cortical) functions – most debates regard domain-specificity, penetrability, encapsulation, innateness

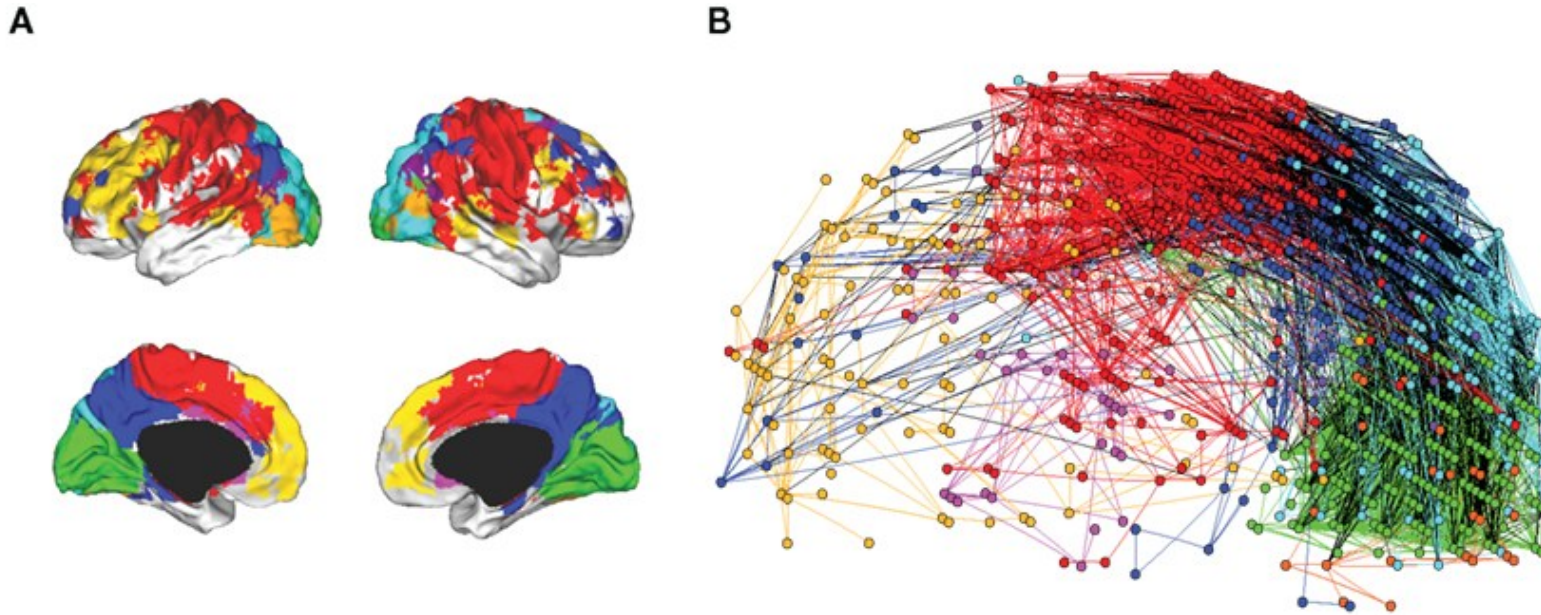
Topological Modularity (network theory)

The brain is hierarchically modular, with **modules** and **sub-modules**.

Modules are defined in terms of topology. They are network communities in a connectome.

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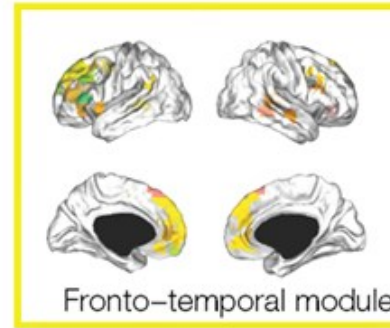
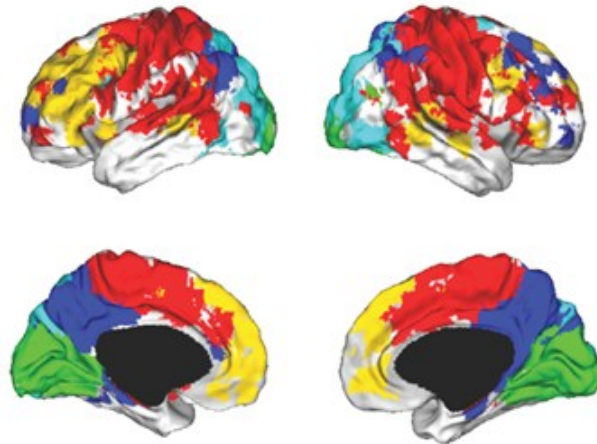
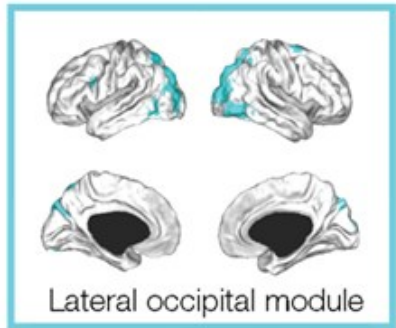
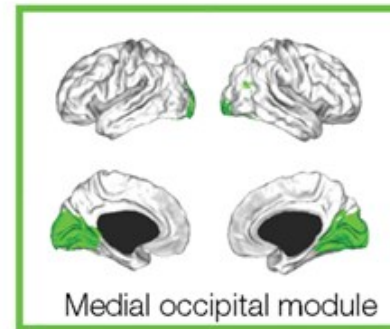
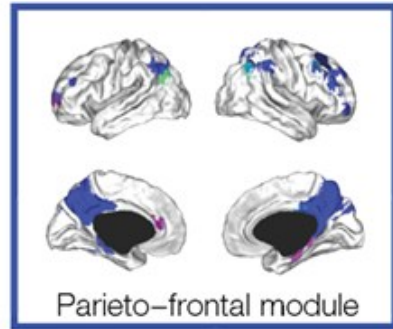
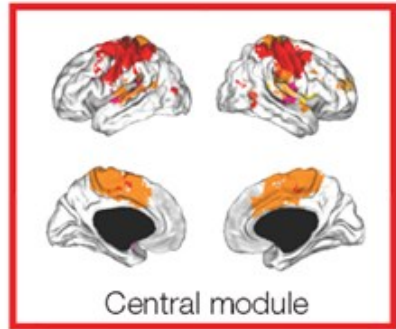


Meunier, Lambiotte et Bullmore, 2010.
Frontiers in Neuroscience

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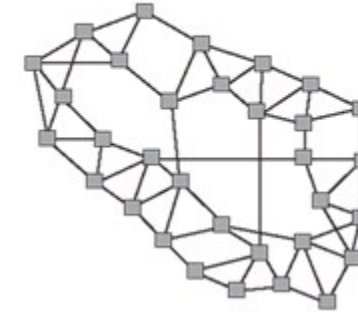
Topological Modularity: Early networks

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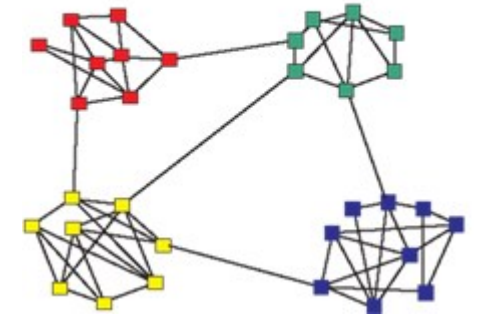


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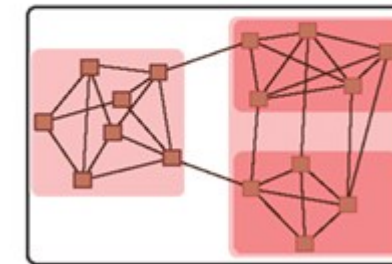
Small-world



Module



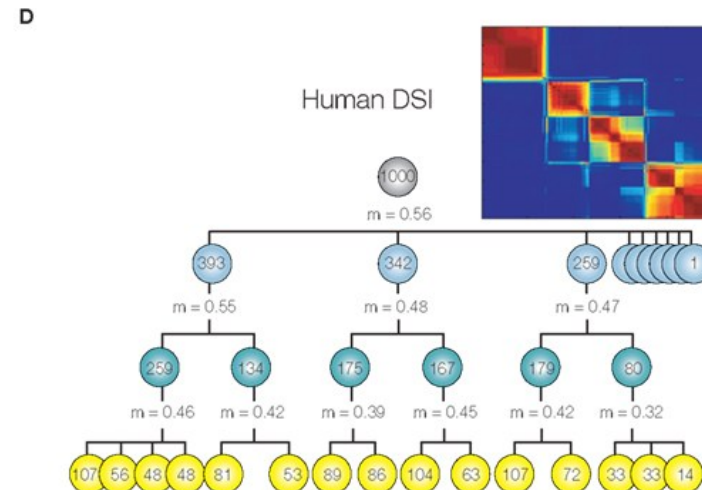
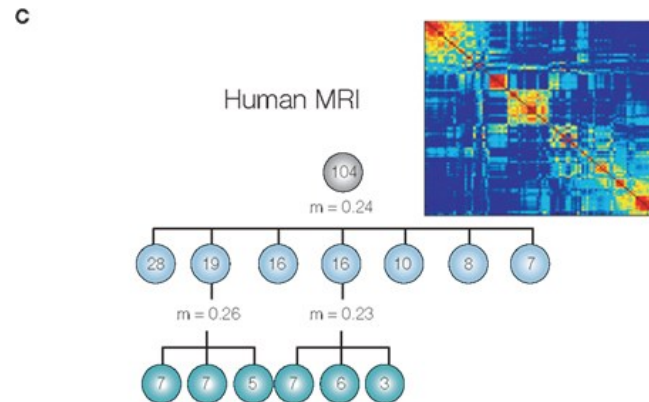
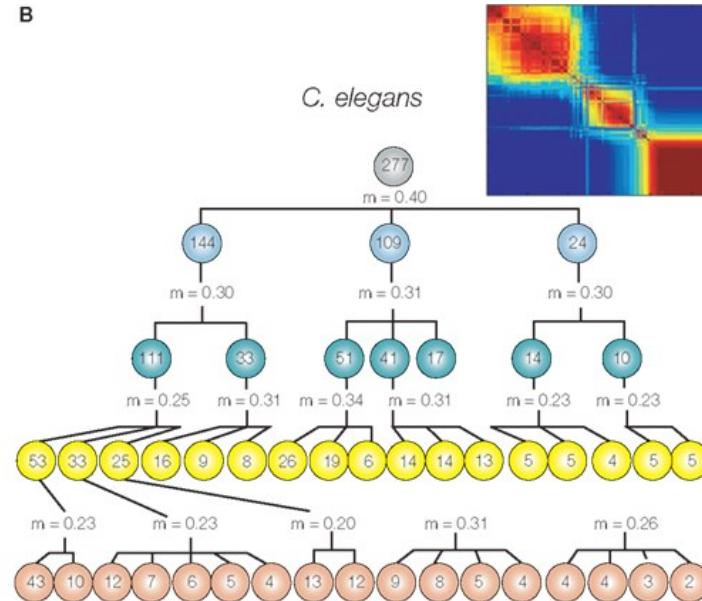
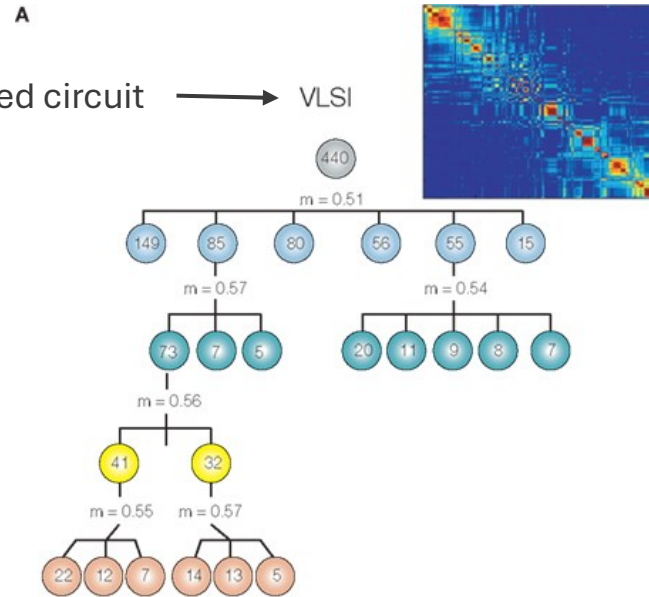
Hierarchy



Module
(specialized function)

Sub-modules
(segregated processes)

Topological Modularity: A property of information-processing systems



« Many systems have the **fractal** property of hierarchical modularity, multi-scale modularity or *russian doll* modularity »

Diffusion spectrum imaging

Topological Modularity: Recent networks

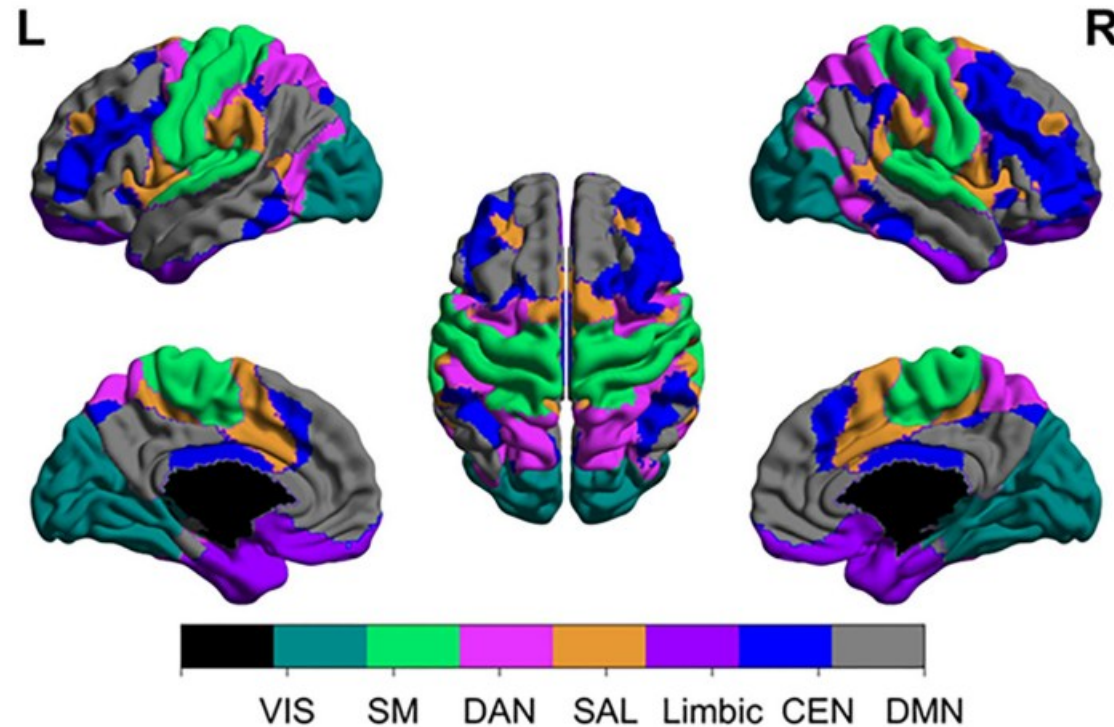


FIGURE 1

Anatomical topographies of canonical large-scale networks. Networks were generated from those reported in Schaefer et al. (2018) (available at: https://github.com/ThomasYeoLab/CBIG/tree/master/stable_projects/brain_parcellation/Schaefer2018_LocalGlobal) with updated terminology. VIS, visual; SM, somatomotor; DAN, dorsal attention network; SAL, salience; CEN, central executive network; DMN, default mode network.

“In recent years, a paradigm shift in neuroscience has been occurring from “**localizationism**,” or the idea that the brain is organized into separately functioning modules, toward “**connectomics**,” or the idea that interconnected nodes form networks as the underlying substrates of behavior and thought.”

Boerger et al., 2023.
Frontiers in Human Neuroscience

Topological Modularity: Modularity of anatomy and functions

Meunier, Lambiotte et Bullmore, 2010. *Frontiers in Neuroscience*
Seguin, Sporns et Zalesky, 2023. *Nature reviews neuroscience*

Cognitive modularity

- Modularity is a property of brain regions
 - Mind organized into **modules**: specialized systems for processing certain inputs (vision, language)
 - Modules defined by: **domain specificity** (operate only on certain inputs), **encapsulation** (insulated from other knowledge), **automaticity** (mandatory operations), relatively **fixed neural architecture**
 - *Psychological-level* definition: modules are computational “boxes” in the mind

Network modularity (predictive coding -compatible)

- Modularity is a property of networks that optimizes information communication
 - A module is a cluster of brain regions with **dense internal connectivity, sparse external connectivity and recursivity**
 - Brain networks are **hierarchically modular**: large modules (e.g., visual network) subdivided into smaller submodules (ventral vs. dorsal visual pathways) subdivided into smaller submodules
 - Modularity provides **computational advantages**: efficiency, robustness, balance between segregation (specialization) and integration (coordination)
 - Modules forward information **with dedicated paths and diffusion processes** to achieve *optimal signalling delay*
-

Why are brain networks expected to be modular?

- **Small-world design:** Favors high clustering within modules supports **locally segregated processing** (e.g., visual motion detection) at low wiring cost, while short path lengths **enable global integration** for generic functions (e.g., working memory)
- **Rich non-linear dynamic behaviors:** Modularity supports **fast intra-modular and slower inter-modular processes**; neural activity can remain **locally encapsulated**; **neural activity is balanced**: *activity doesn't die, doesn't spread network-wide*; **redundancy of signal**; **flexibility and stability** by allowing modules and submodules **reconfiguration**
- **Origins:** Modular networks arise naturally when structure and dynamics co-evolve. Connections between regions that tend to synchronize are reinforced; weak or unsynchronized links are pruned. Specialized submodules can be reused

End quote

“In the psychological literature, the central principle of phrenology or faculty psychology has been that mental function can be somehow sub-divided into part-functions or mental modules (Fodor, 1983).

Modular processes, like color vision, have been described as automatic, effortless, informationally encapsulated, and anatomically localized (Zeki and Bartels, 1998).

More consciously effortful tasks, like working memory, have been proposed to demand access to a more globally integrated processing system – a workspace of synchronized neurons oscillating coherently over large physical distances across the whole brain (Varela et al., 2001; Buzsáki and Draguhn, 2004).

In short, there are strong prior reasons to believe that brain networks are formed and function as modular systems.”
